



IEEE Recommended Practice for Techno-Economic Metrics for Grid Power Generators and Energy Storage Systems

IEEE Systems, Man, and Cybernetics Society

Developed by the
Standards Committee

IEEE Std 2814™-2025

STANDARDS

IEEE Recommended Practice for Techno-Economic Metrics for Grid Power Generators and Energy Storage Systems

Developed by the

Standards Committee
of the
IEEE Systems, Man, and Cybernetics Society

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IEEE SA Standards Board

Abstract: Techno-economic terminologies used in the construction and operation of low-carbon power generation and energy storage systems are defined. Techno-economic terminologies lack consistent definitions in today's evolving low-carbon power generation and electrical energy storage. Stakeholders will be supported in using a common language when planning standards and for understanding published industrial and technical reports.

Keywords: energy storage systems, financial model, grid power generators, IEEE 2814, techno-economic

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Introduction

This introduction is not part of IEEE Std 2814-2025, IEEE Recommended Practice for Techno-economic Metrics for Grid Power Generators and Energy Storage Systems.

This document provides support on how to perform techno-economic analysis for grid power generators and energy storage systems. The recommended practice defines a set of cost and finance metrics to benchmark and evaluate the techno-economic performance of energy systems. This recommended practice is intended to serve as a basic reference for policymakers, system developers, and users of such systems, for planning industry standards, and for the interpretation of published technical and industrial reports.

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IEEE Recommended Practice for Techno-Economic Metrics for Grid Power Generators and Energy Storage Systems

1. Overview

1.1 General

This recommended practice is organized as follows: Clause 1 provides an overview of the scope of this recommended practice. Clause 2 provides the normative references. Clause 3 provides definitions that are either unavailable in the IEEE Standards Dictionary Online or have been modified for use with this standard. Clause 4 presents a justification for this proposal with a brief description. Clause 5 introduces the methodology to assess the techno-economic performance of power generation and energy storage systems, which is detailed in the three subsequent clauses. Namely, Clause 6 describes the levelized cost of energy analysis methodologies that will be considered for conducting the cost analysis. Clause 7 describes the financial analysis methodologies.

Annexes are included for completeness. Annex A provides examples of applications of the described methodology for conducting techno-economic analysis for wind and solar photovoltaic technologies combined with energy storage systems. Annex B provides the repository data. Annex C provides a bibliography.

1.2 Scope

This recommended practice describes the techno-economic terminologies used in the construction and operation of low-carbon energy generation and energy storage systems. The scope addresses how to perform techno-economic analysis for grid power generators and energy storage systems. The recommended practice defines a set of cost and finance metrics to benchmark and evaluate the techno-economic performance of energy systems. Energy storage systems providing backup power are not within the scope.

1.3 Purpose

There are no consistent definitions of the techno-economic terms that have arisen in the evolution of low-carbon energy generation and energy storage systems. There is a need to define this techno-economic terminology so that stakeholders can use a common language when planning standards and for understanding published industrial and technical reports. This standard is intended to serve as a basic reference for policymakers, developers, and users of such systems, for planning industry standards, and for the interpretation of published technical and industrial reports.

1.4 Word usage

The word *shall* indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (*shall* equals *is required to*).^{6, 7}

The word *should* indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should* equals *is recommended that*).

The word *may* is used to indicate a course of action permissible within the limits of the standard (*may* equals *is permitted to*).

The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can* equals *is able to*).

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 1561™-2019, IEEE Guide for Optimizing the Performance and Life of Lead-Acid Batteries in Remote Hybrid Power Systems.^{8, 9}

IEEE Std 2030.9™-2019, IEEE Recommended Practice for the Planning and Design of the Microgrid.

ISO/IEC 13273-2:2015(en), Energy efficiency and renewable energy sources—Common international terminology—Part 2: Renewable energy sources.¹⁰

3. Definitions, acronyms, and abbreviations

3.1 Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.¹¹

balance of system (BOS): All equipment, other than the main components for generation, storage, or power conversion, needed to support the delivery of energy.

contract for difference (CfD): A low-carbon electricity generation investment incentive, providing predictability and stability to revenue streams.

⁶ The use of the word *must* is deprecated and cannot be used when stating mandatory requirements; *must* is used only to describe unavoidable situations.

⁷ The use of *will* is deprecated and cannot be used when stating mandatory requirements; *will* is only used in statements of fact.

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¹⁰ ISO publications are available from the International Organization for Standardization (<https://www.iso.org/>) and the American National Standards Institute (<https://www.ansi.org/>).

¹¹ *IEEE Standards Dictionary Online* is available at: <http://dictionary.ieee.org>. An IEEE account is required for access to the dictionary, and one can be created at no charge on the dictionary sign-in page.

cost of debt: The effective rate that a company pays on its debt, such as bonds and loans.

cost of equity: The return that a company requires for an investment or project, or the return that an individual requires for an equity investment.

debt: A liability that has to be paid off after a period with interest.

debt duration: The amount of time for the project to repay the debt to the debt holders.

discounted cash flow (DCF): An appraisal methodology to assess the economic and financial investment indicators based on cash flows (positive and negative) incurred over the project's life cycle.

earnings before interest and taxes (EBIT): A measure of the profit that includes all incomes and expenses, without income tax expenses and interest expenses.

equity: The cash raised by the company by issuing shares representing units of ownership.

generation integrated energy storage (GIES): A class of energy storage storing energy during the transformation between the primary energy form and electricity.

levelized cost of delivery (LCOD): The average cost of delivering electricity, including generation, transmission, and distribution, over the lifetime of an energy system.

levelized cost of energy (LCOE): The average cost of electricity produced by a generator over its lifetime.

levelized cost of storage (LCOS): The average cost of electricity discharged by an energy storage system over its lifetime.

NPV to the equity: This is the sum of the levered cash flows. Discounting the “free cash flow to the equity” at the cost of equity gives the net present value (NPV) to the equity.

NPV to firm: Net present value (NPV) to firm is the sum of the unlevered cash flows discounted with the weighted average capital cost (WACC) or the “free cash flow to firm.”

techno-economic analysis: Examination of costs, benefits, risks, uncertainties, and timeframes to evaluate the attributes of energy technologies.

unit of analysis: The entity that frames what is being looked at in a study, or is the entity being studied as a whole.

weighted average capital cost (WACC): The rate that a company is expected to pay, on average, to all its security holders to finance its assets.

3.2 Acronyms and abbreviations

BOS	balance of system
CAES	compressed air energy storage
CAPEX	capital expenditure
CfD	contract for difference
DCF	discounted cash flow

EBIT	earnings before interest and taxes
FCFE	free cash flow to equity
FCFF	free cash flow to firm
FR	fast reserve
GIES	generation integrated energy storage
IRR	internal rate of return
LCOD	levelized cost of delivery
LCOE	levelized cost of energy/electricity
LCOS	levelized cost of storage
NPV	net present value
O&M	operation and maintenance
PDF	probability distribution function
PSH	pumped storage hydropower
PTES	pumped thermal energy storage
STOR	short-term operating reserve
VRB	vanadium redox battery
WACC	weighted average capital cost

4. Need for the recommended practice

4.1 General

Power systems are experiencing a rapid transition on how electricity is generated and consumed, driven by decarbonization and sustainability. Low-carbon power generation technologies, including intermittent renewables (e.g., wind and solar) and nuclear power, will be installed in various locations and capacities as defined by ISO/IEC 13273-2:2015.¹² The transition from the traditional power systems to smart grids is inevitable. One of the key challenges with adopting low-carbon generation technologies is the issue of dispatchability. For example, wind and solar power generation are proportional to the wind and solar resources available. Nuclear power generation is normally operated as a fixed output and cannot be ramped up or down. Hence, energy storage will play a key role in providing smart grids operational flexibility to balance power generation and demand.

Grid energy storage can store surplus energy (e.g., from renewable generation) for later use when required. There are several types of energy storage systems, including mechanical-based (compressed air, pumped storage hydropower, flywheel), electrical-based (supercapacitors and superconducting magnetic), electrochemical-based (lead-acid, lithium-ion, and flow batteries), and hydrogen energy storage. Figure 1

¹² Information on references can be found in Clause 2.

provides an overview of energy transformation processes between low-carbon power generation and common energy storage technologies.

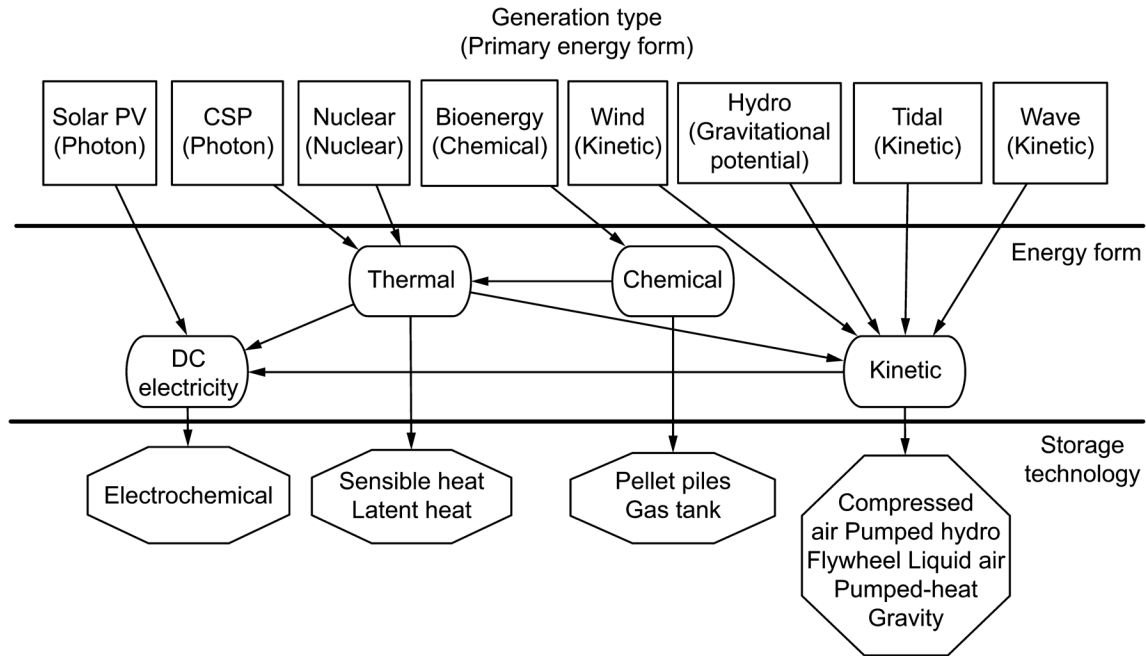


Figure 1—Energy transformation processes between low-carbon power generation and possible energy storage systems (Lai, et al. “A review” [B12])

With the increasing technological maturity and economies of scale for renewable power generator and electrochemical energy storage technologies, there is a potential for mass-scale deployment of both technologies in standalone off-grid and grid-connected power systems.

The challenge arises in analyzing the economic projections on complex energy systems utilizing low-carbon power generator and energy storage technologies. There is no universal and standardized economic appraisal technique to compare the economic viability of various energy generation and storage technologies. This recommended practice provides a standardized way to resolve the aforementioned challenge. The methodologies proposed may not be appropriate for comparing projects of different capital cost requirement characteristics for comparable system electrical characteristics.

4.2 A simple example of application

A simple example will be used to demonstrate when the recommended practice may be applicable for techno-economic appraisal of energy generation and storage technologies that form a microgrid as defined in IEEE Std 2030.9™-2019. Figure 2 shows the energy flow diagram of the renewable energy and energy storage system. The energy output of the power generator is separated into two streams. The net energy output of the energy storage system needs to take into account storage efficiency. The energy storage experiences the energy flowing in and flowing out.

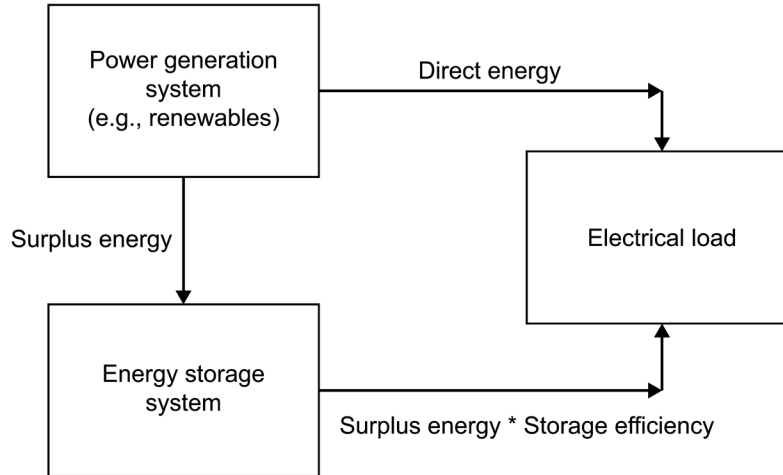


Figure 2—Energy flow diagram of the combined low-carbon power generation and energy storage system

With the availability of energy meters and computational resources, it is viable to obtain high resolution energy data to conduct techno-economic appraisal. The pertinent question that arises in conducting techno-economic appraisal is how to account for the energy and monetary flow in a simple and transparent way that accurately assesses the cost of power generation and energy storage technologies, as well as the overall energy system.

Three key parameters should be considered when combining power generation and energy storage systems. These are the storage (i.e., primary energy form to storage energy form) efficiency (η_S), transmission (i.e., primary energy form to electricity) efficiency (η_T), and the throughput (to examine the overall power generation and energy storage efficiency) efficiency (η_X), defined as follows (Garvey et al. [B4]):

$$\eta_S = \frac{\text{electricity output from the system if all energy passed through storage}}{\text{electricity output from the system if no energy passed through storage}} \quad (1)$$

$$\eta_T = \frac{\text{electricity output from the system if no energy passed through storage}}{\text{total primary energy input to the system}} \quad (2)$$

$$\eta_X = \frac{\text{total electricity output from the system}}{\text{total primary energy input to the system}} \quad (3)$$

4.3 Outline of the recommended practice

This recommended practice provides the fundamental information regarding the description of possible techno-economic appraisal practices for power generation and energy storage technologies in power systems. The following clauses present the approach. Clause 5 introduces the methodology to assess the techno-economic performance of power generation and energy storage systems, which is detailed in the three subsequent clauses. Namely, Clause 6 describes the levelized cost of energy analysis methodologies that will be considered for conducting the cost analysis. Clause 7 describes the financial analysis methodologies. Annex A provides examples of the application of the described methodology for conducting techno-economic analysis for wind and solar photovoltaic technologies combined with energy storage systems. Annex B provides the repository data. Annex C provides a bibliography.

5. Methodology to assess the techno-economic performance of power generators and energy storage systems

5.1 General

There is no unique practice that solves all techno-economic appraisal problems, considering the possible application scenarios. The suitability of a practice is highly dependent on the application-specific characteristics and significances. Therefore, the objective is to establish a methodology to perform techno-economic appraisal for a given application scenario as suggested by the user. As the energy flow and monetary flow of energy technologies are complicated in nature, the user should first define the unit of analysis for a particular energy technology. This could be the energy storage system, the power generator, or the whole energy system.

A discounted cash flow (DCF) analysis is an appraisal methodology to assess the economic and financial investment indicators based on cash flows (positive and negative) incurred over the project's life cycle. A DCF analysis establishes the present value of expected future cash flows with a discount rate. The total value of all the DCFs (costs and revenues) is called the *net present value (NPV)*. Generally, the investment needs a positive NPV to sufficiently remunerate debt and equity holders.

Figure 3 shows the schematic and key relationships between stakeholders within the financial model. The typical financing resources include debt and equity. *Debt* is the company's liability and has to be paid off after a period with interest. *Equity* is the cash raise by the company by issuing shares representing units of ownership. *Earnings before interest and taxes (EBIT)* measures the profit that includes all incomes and expenses, without income tax expenses and interest expenses. Equity holders expect a return from the dividend, which is variable and depends on the business performance.

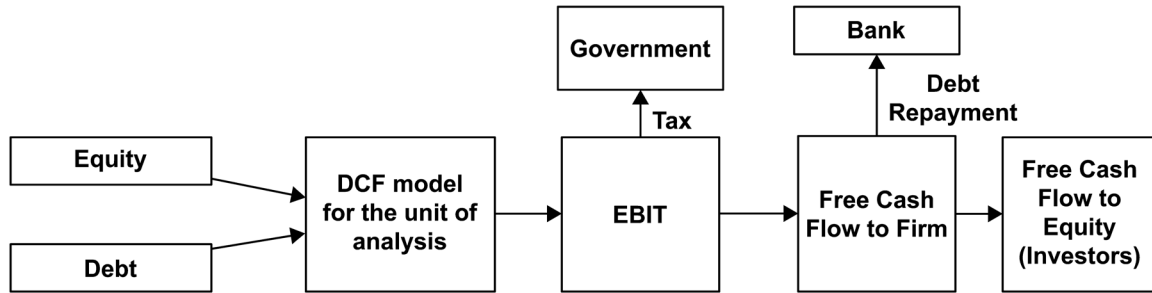


Figure 3—Methodology for techno-economic analysis of power generators and energy storage systems (Lai and Locatelli “Economic and financial” [B10])

The DCF model should consider three outputs:

- **Levelized cost of electricity (LCOE):** This is the price of the electricity necessary to cover all the life cycle cost. The formulation of LCOE is:

$$\text{LCOE} = \frac{\sum_{n=0}^N \frac{C_{\text{cap}_n} + C_{\text{O\&M}_n}}{(1 + \text{WACC})^n}}{\sum_{n=0}^N \frac{E_n}{(1 + \text{WACC})^n}} \quad (4)$$

where

C_{cap} is the capital cost (\$)

$C_{\text{O\&M}}$ is the operational and maintenance cost (\$)

E is the system energy output (kWh)

N is the system lifetime (years)

Generally, the cash flow for the LCOE is discounted at the discount rate or the weighted average cost of capital (WACC) (U.S. Energy Information Administration [EIA] [B16]). The WACC calculation is given as follows (Lai et al. “A financial model” [B11]):

$$\text{WACC} = K_E \times \theta_{\text{CAPEX}} + K_D \times (1 - \theta_{\text{CAPEX}}) \times (1 - \theta_{\text{Tax}}) \quad (5)$$

where

K_E is the cost of equity (%)

K_D is the cost of debt (%)

θ_{CAPEX} is the equity share on CAPEX (%)

θ_{Tax} is the effective tax rate (%)

The LCOE provides a holistic perspective and is used by policymakers and engineers to compare technologies and estimate the electricity sales price for the technology to break-even.

In the financial analysis and investment appraisal, there are, therefore, two different relevant NPV.

- **NPV to firm:** This is the sum of the unlevered cash flows discounted with the WACC or the “free cash flow to firm.” Debt holders expect remuneration regardless of NPV and receive remuneration if the NPV is greater than zero. This NPV accounts for the tax rate and earnings before interest. NPV to firm is based on the cash flows before deducting the financial obligation (e.g., interest and debt payments).
- **NPV to the equity:** This is the sum of the levered cash flows. Discounting the “free cash flow to the equity” at the cost of equity gives the NPV to the equity. It is the NPV from the perspective of equity holders (i.e., the “owner of the energy storage system”) after the debt has been repaid to the debt holders. The equity holders receive a payment equal to the cost of equity if the NPV is zero.

The inputs for the DCF model can be categorized into three major categories, namely, technical, economic, and financial. Figure 4 presents the financial and economic modelling process. The technical inputs are classified into *power and energy rating*, *efficiency parameters and power ratios*, and *revenue*. The economic inputs can be classified as *system-specific* (i.e., fixed and variable costs of sub systems) and *revenue* (e.g., price of electricity).

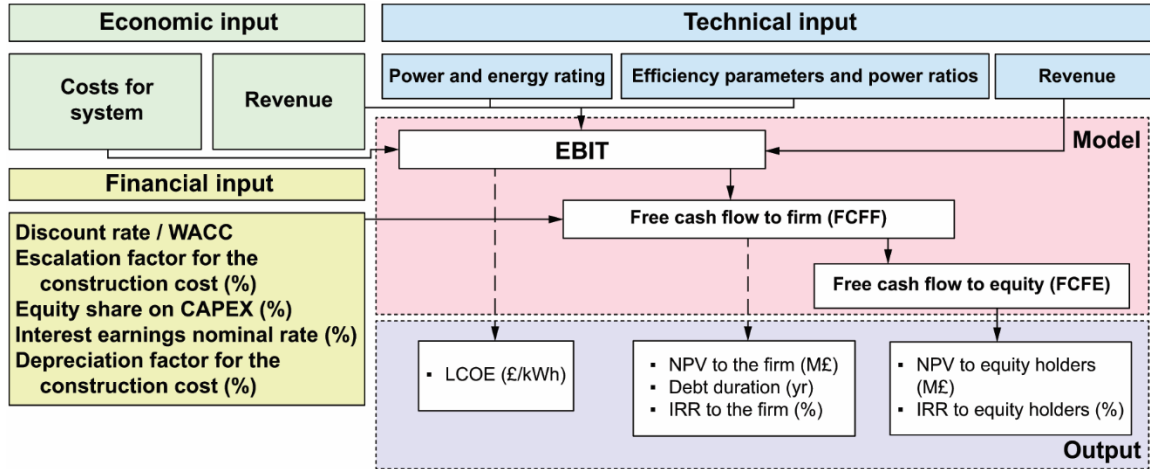


Figure 4—Technical, financial, and economic inputs for combined power generation and energy storage systems techno-economic analysis

The details of the techno-economic analysis, as shown in Figure 4, are given below.

5.2 Costs for system (economic input)

This input should be concerned with the capital, fixed and variable operation, and maintenance costs associated with the power generation and energy storage technologies, such as:

- System lifetime (excluding construction) (years)
- Construction time (years)
- Specific fixed operation and maintenance (O&M) power cost for generator (k\$/MW-year)
- Specific fixed O&M power cost for energy storage system (\$/MW-year)
- Specific generator overnight cost (\$/MW); the unit of measure considers power volume as the generator mainly delivers power
- Specific balance of system (BOS) for generator cost (k\$/MW)
- Specific BOS for energy storage cost (k\$/MWh)
- Specific energy storage overnight cost (k\$/MWh); the unit of measure considers energy volume as energy storage mainly stores and delivers energy for a given period (Cole and Karmakar [B2])
- Annual inflation rate for cash (%)

Overnight cost is the cost of a construction project if no interest was incurred during construction, as if the project were completed “overnight.”

5.3 Revenue (economic input)

This input should be concerned with the monetary reward (\$) given to the power generation and energy storage technologies during commitment to capacity/ancillary service programs, such as the National Energy System Operator [NESO] [B14]):

- Wholesale energy market
- Frequency response services

- Reserve services
- System security services
- Reactive power services
- Balancing mechanisms

A comprehensive list of revenue streams for energy storage systems can be found here (Lai and Locatelli “Are energy policies” [B9]).

5.4 Power and energy rating (technical input)

This input should be concerned with the power and energy rating associated with the power generation and energy storage technologies. The resource availability (i.e., capacity factor) and the system’s performance degradation should be considered. Such inputs include:

- Power rating for recovering energy from storage (MW)
- Power rating for putting energy into storage (MW)
- Power rating for electricity conversion (MW)
- Primary harvester power rating (MW)
- ES energy capacity (MWh)
- ES power capacity (MW)
- ES output at year 1 with renewable generator energy only (MWh/yr)
- Total storage energy output at year 1 (MWh/yr); this is all the energy that is delivered by energy storage with energy input from all kinds of sources, e.g., grid
- Storage degradation (%/year)
- Primary source energy output (MWh/year)
- Capacity factor
- Power rating committed to capacity/ancillary service program (MW)

5.5 Efficiency parameters and power ratios (technical input)

This input should be concerned with the efficiency of power generation and energy storage systems. The energy flow to and out of energy storage should be accounted for. The power ratio should determine the sizing of BOS for energy conversions. The storage BOS includes the supporting cost components for the storage block, including container, cabling, switchgear, flow battery pumps, and heating, ventilation, and air conditioning (HVAC) (Pacific Northwest National Laboratory [B15]). The power input from the primary energy source should be taken as the reference rating, acting as the denominator for each of the three ratios. Such inputs include:

- Storage (round-trip) efficiency (%)
- Transmission efficiency (%)
- Throughput efficiency (%)
- Fraction of electrical energy output passed through storage (%)
- Fraction of primary electrical energy input that will pass through storage (%)

- Power ratio for recovering energy from storage
- Power ratio for putting energy into storage
- Power ratio for electricity generation

5.6 Revenue (technical input)

This input should be concerned with the energy supplied and time committed to capacity/ancillary service programs by the system. Such inputs include:

- Total hours availability commitment to capacity/ancillary service program (hours/year)
- Capacity/ancillary service program annual energy utilization (MWh/year)
- Energy supplied to wholesale market (MWh/year)

5.7 Discount rate/WACC (financial input)

This input should be concerned with the financial parameters associated with project investments. Such inputs include:

- Percentage of equity (%)
- Percentage of debt (%)
- Market value of equity (\$)
- Market value of debt (\$)
- Cost of equity (%)
- Cost of debt (%)
- Corporate tax rate (%)

6. Levelized cost of energy analysis methodologies

6.1 General

Cost comparison is one of the basic requirements for techno-economic analysis. As power generation and energy storage systems are inherently different, the LCOE for individual components and the combined system can be studied in detail.

The user should consider the energy flows and their cost implications for power generation and energy storage systems. As described in Clause 5, the cost is largely affected by the asset lifetime electricity production, as well as the operational and degradation costs. The original definition of LCOE is given in Equation (4).

The following clause will demonstrate how LCOE should be appraised for typical power generation (i.e., dispatchable and intermittent), energy storage, and the overall energy system (Lai and McCulloch [B13]). For energy storage, the analogous terms levelized cost of storage (LCOS, see 6.4) and levelized cost of delivery (LCOD, see 6.5) apply.

6.2 Dispatchable power generator

This type of power generator is concerned with dispatchable sources, including anaerobic digestion, biogas, and biomass power plants. The LCOE for a dispatchable power generator can be calculated with Equation (6), Equation (7), Equation (8), and Equation (9).

$$\text{LCOE}_{AD} = \frac{C_{AD}}{E_{AD}} \quad (6)$$

$$C_{AD} = \sum_{i=0}^n \frac{C_{Cap_{AD}}(i) \times P_{AD_{Max}} + C_{AD_{O\&M_{Fixed}}} \times P_{AD_{Max}} + \sum_{j=1}^t C_{AD_{O\&M_{Var}}}(i, j)}{(1 + \text{WACC})^i} \quad (7)$$

where

$C_{Cap_{AD}}$ is the capital annual costs for the generator at hour j and year i

$C_{AD_{O\&M_{Fixed}}}$ is the fixed annual O&M costs for the generator at hour j and year i

$C_{AD_{O\&M_{Var}}}$ is the variable operation and maintenance cost for the generator, calculated with Equation (8)

$$C_{AD_{O\&M_{Var}}}(i, j) = C_{AD_{Fuel}}(i, j) + C_{AD_{Labor}}(i, j) \quad (8)$$

where

$C_{AD_{Labor}}$ is the labor cost (\$/kWh) for operating the generator

$C_{AD_{Fuel}}$ is the fuel cost (\$) for operating the generator

The generator lifetime energy production at present value is calculated with Equation (9):

$$E_{AD} = \sum_{i=0}^n \frac{\sum_{j=1}^t P_{AD_{Direct}}(i, j) + P_{AD_{Store}}(i, j)}{(1 + \text{WACC})^i} \quad (9)$$

where

$P_{AD_{Direct}}$ is the generator power output that is used to meet the load directly

$P_{AD_{Store}}$ is the surplus power from generator that stored in energy storage system for later use

6.3 Intermittent power generator

This type of power generator is concerned with intermittent sources, including wind and solar power generators. The LCOE for the intermittent power generator can be calculated with Equation (10), Equation (11), and Equation (12).

$$\text{LCOE}_{IG} = \frac{C_{IG}}{E_{IG}} \quad (10)$$

$$C_{IG} = \sum_{i=0}^n \frac{C_{Cap_{IG}}(i) + C_{Inst_{IG}}(i) + C_{IG_{O\&M_{Fixed}}}}{(1 + \text{WACC})^i} \quad (11)$$

$$E_{IG} = \sum_{i=0}^n \frac{(1 - \sigma)^i \times \sum_{j=1}^t P_{IG_{Direct}}(i, j) + P_{IG_{Store}}(i, j)}{(1 + \text{WACC})^i} \quad (12)$$

where

- $C_{Cap_{IG}}$ is the capital cost for the generator
- $C_{Inst_{IG}}$ is the installation cost for the generator
- $C_{IG_{O\&M_{Fixed}}}$ is the operation and maintenance cost for the generator
- σ is the annual degradation constant for the intermittent generator (e.g., photovoltaic panels)
- $P_{IG_{Direct}}$ is the power output from the generator that is used to meet the load at hour j and year i
- $P_{IG_{Store}}$ is the power output from the generator that is to be stored in energy storage, at hour j and year i

For an off-grid system, the curtailed energy from intermittent generator, i.e., not used directly to meet the load or stored in energy storage, should not be included in the lifetime energy production as the generated electricity will not be sold and the cost cannot be recovered, resulting a higher LCOE.

6.4 Energy storage

Levelized cost of storage (LCOS) is the analogous term for LCOE used for power generators. LCOS has similar attributes to LCOE and is given by Equation (13) below:

$$\text{LCOS} = \frac{C_{EES}}{E_{EES}} \quad (13)$$

$$C_{EES} = \sum_{i=0}^n \frac{C_{Cap_{EES}}(i) + C_{EES_{O\&M_{Total}}}}{(1 + \text{WACC})^i} \quad (14)$$

$$C_{EES_{O\&M_{Total}}} = C_{EES_{O\&M_{Fixed}}} + \sum_{k=1}^m C_{EES_{DegkWh}}(i, k) + \sum_{j=1}^t C_{EES_{kWh}}(i, j) \quad (15)$$

where

- $C_{EES_{O\&M_{Fixed}}}$ is the fixed annual EES O&M cost
- m is the number of discharge cycles per year

The variable operational costs for the energy storage system include the degradation per kWh cost, and energy costs are calculated with Equation (16) and Equation (17), respectively.

$$C_{EES_{DegkWh}}(i, k) = \frac{C_{Cap_{EES}}}{Ratedcycle_{SOC_{Upper}, SOC_{Lower}}(i, k)} \times [SOC_{Upper}(i, k) - SOC_{Lower}(i, k)] \times E_{EES_{Rated}} \quad (16)$$

$$C_{EES_{kWh}}(i, j) = C_{EES_{O\&M_{Var}}} \times E_{EES_{Store}}(i, j) \quad (17)$$

where

$C_{EES_{O\&M_{Var}}}$ is the variable O&M cost for the energy $E_{EES_{Store}}$ to be stored in the energy storage system

The lifetime electricity production is calculated with Equation (18).

$$E_{EES-life} = \eta_{EES} \sum_{i=0}^n \frac{\sum_{j=1}^t E_{EES}(i, j)}{(1 + WACC)^i} \quad (18)$$

where

η_{EES} is the round-trip efficiency of the energy storage system

6.5 Levelized cost of delivery (LCOD)

Since the energy storage system is not a conventional electrical energy generating source, i.e., the asset does not generate electricity from the primary energy source, the user should consider the cost for the energy conversion process from the primary form, e.g., solar irradiance or biomass into electricity (Lai and McCulloch [B13]). Equation (19) gives the LCOD for the system.

$$LCOD = \frac{C_{EES} + C_{Gen} + C_{Con}}{E_{EES-life}} \quad (19)$$

The charge controller cost is given in Equation (20).

$$C_{Con} = N_{Con} \times (C_{Cap_{Con}} + C_{InstCon}) + \sum_{i=0}^n \frac{C_{O\&M_{Con}} \times N_{Con}}{(1 + d)^i} \quad (20)$$

$$N_{Con} = ceil\left(\frac{P_{PV_{Rated}}}{P_{Con}}\right) \times N_{Con-rep} \quad (21)$$

where

N_{Con} is the number of controllers

$C_{InstCon}$ is the capital and installation costs for the controller; the land leasing cost should be accounted for in the capital cost

$C_{O\&M_{Con}}$ is the O&M costs for the controller; the ceil function converts the number of controllers into the nearest integer above

$N_{Con-rep}$ is the number of controller replacements during the system's lifetime

$P_{PV_{Rated}}$ is the rated power output for the intermittent power generators

P_{Con} is the rated power output for the controller

There are reasons to charge the energy storage system with a dispatchable power generator as defined in IEEE Std 1561™-2019. This could be to enhance power system energy security or to improve the system's performance (e.g., some generators, including biogas power plants, have poor part-load efficiency). The cost for producing the electricity from dispatchable or intermittent power generators to be stored in the energy storage system is given in Equation (22) and Equation (23).

$$C_{Gen} = \sum_{i=0}^n \frac{C_{Cap_{AD}}(i) + C_{AD_{O\&M_{Fixed}}} + \sum_{j=1}^t C_{AD_{O\&M_{Store}}}(i, j)}{(1+d)^i} \quad (22)$$

$$C_{AD_{O\&M_{Store}}}(i, j) = C_{AD_{Fuel}}(i, j) + C_{AD_{Labor}} \times P_{AD_{Store}}(i, j) \quad (23)$$

where

$P_{AD_{Store}}$ is the power generated from the power generator to be stored in the energy storage system as energy

6.6 LCOE of the overall energy system

The system's LCOE is derived in Equation (24). It takes into account the total costs for operating the system and the energy output to meet the load.

$$LCOE_{system} = \frac{C_{PV} + C_{EES} + C_{Con} + C_{Inv} + C_{AD}}{E_{EES} + E_{Gen_{Directtotal}}} \quad (24)$$

$$C_{Inv} = \sum_{i=0}^n \frac{N_{Inv} \times (C_{Cap_{Inv}}(i) + C_{Inst_{Inv}}(i)) + C_{O\&M_{Inv}}(i) \times N_{Inv}}{(1+d)^i} \quad (25)$$

where

$C_{Cap_{Inv}}$ is the capital cost for the bi-directional inverter

$C_{Inst_{Inv}}$ is the installation cost for the bi-directional inverter

$C_{O\&M_{Inv}}$ is the O&M cost for the bi-directional inverter

The number of required inverters is calculated with Equation (26) below:

$$N_{inv} = \text{ceil} \left(\frac{P_{PV_{Rated}}}{P_{Inv}} \right) \times N_{Inv-rep} \quad (26)$$

where

$P_{PV_{Rated}}$ is the rated capacity of generator

$N_{Inv-rep}$ is the number of inverter replacements during the system's lifetime

The total energy produced by the power generator during system operation that is used to meet the load directly are given in Equation (27).

$$E_{Gen_{Directtotal}} = \sum_{i=0}^n \frac{(1-\sigma)^i \times \sum_{j=1}^t P_{Gen_{Direct}}(i, j)}{(1+d)^i} \quad (27)$$

where

$P_{Gen_{Direct}}$ is the power generated from power generator to meet the load directly

7. Financial analysis methodologies

The methodology, as shown in Clause 5, determines the NPVs, debt durations, and internal rates of return (IRRs) with free cash flow to firm (FCFF) and free cash flow to equity (FCFE). The LCOE is calculated with the EBIT cash flow. Equation (28) and Equation (29) present the NPV and IRR to the firm. Equation (30) and Equation (31) present the NPV and IRR to the equity. The equity NPV needs to be discounted at the cost of equity and not the WACC.

$$\text{NPV to firm} = \sum_{t=0}^n \frac{\text{Cash flow to firm}_t}{(1 + \text{WACC})^t} \quad (28)$$

$$\sum_{t=0}^n \frac{\text{Cash flow to firm}_t}{(1 + \text{IRR}_{\text{firm}})^t} = 0 \quad (29)$$

$$\text{NPV to the equity} = \sum_{t=0}^n \frac{\text{Cash flow to the equity}_t}{(1 + K_E)^t} \quad (30)$$

$$\sum_{t=0}^n \frac{\text{Cash flow to the equity}_t}{(1 + \text{IRR}_{\text{equity}})^t} = 0 \quad (31)$$

The debt duration is the amount of time for the project to repay the debt to debt holders. This is calculated as

$$\sum_t \text{Debt}_{\text{Duration}}(t).$$

where

D_{Cum_t} is the debt cumulated (M\$)

E_{Cum_t} is the equity cumulated (M\$)

The debt duration is calculated with Equation (32) and Equation (33).

$$\text{Debt}_{\text{Duration}}(t) = \begin{cases} 1, & \text{if } \text{Debt}_{\text{Percentage}}(t) > 0 \\ 0, & \text{otherwise} \end{cases} \quad (32)$$

$$\text{Debt}_{\text{Percentage}}(t) = \begin{cases} \frac{D_{\text{Cum}_t}}{D_{\text{Cum}_t} + E_{\text{Cum}_t}}, & \text{if } D_{\text{Cum}_t} > 0 \text{ and } E_{\text{Cum}_t} > 0 \\ 1, & \text{if } E_{\text{Cum}_t} \leq 0 \\ 0, & \text{otherwise} \end{cases} \quad (33)$$

Annex A

(informative)

Examples of application of the methodology

A.1 Techno-economic analysis for wind power generator and energy storage in the United Kingdom

The following example is reproduced from Lai and Locatelli, “Economic and financial appraisal of novel large-scale energy storage technologies” [B10].

In the United Kingdom, wind power alone accounts for 21.2% of the UK’s total electricity generation in 2019 and is by far the most relevant non-dispatchable source of power. Therefore, this example will present the techno-economic analysis for a wind power generator with two energy storage options: generation-integrated energy storage (GIES, with pumped heat energy storage) and non-GIES (with Li-ion battery).

Figure A.1 presents the energy flow of a GIES, non-GIES, and wind-only systems concerning the revenue sources. Four revenue sources that may be applicable for wind power and energy storage systems are the Contract for Difference (CfD), wholesale market, short-term operating reserve (STOR), and fast reserve (FR).¹³

The key hypothesis of this example is:

- a) The power and energy rating for the wind power generator (multiple wind turbines) and energy storage system are 100 MW and 100 MWh, respectively.
- b) The wind-only system sells all the energy to the grid at CfD price.
- c) The wind plant with an energy storage system can either sell to the grid at CfD price or store the energy in the energy storage system. To address the power dispatch issue, if there is storage space available in the energy storage system, then the energy is stored. If there is no space, then the energy is sold through the CfD.
- d) The energy in the energy storage system can be sold to the wholesale market or used for grid services.
 - 1) Regarding the wholesale market:
 - i) If the wind power generator cannot provide all the energy for the energy storage system and the energy storage system’s “empty capacity” is available, then the energy storage system will buy electricity from the grid at an average low price (i.e., when there is low demand for electricity).
 - 2) Regarding the grid services:
 - i) The energy storage system can serve both the STOR and the FR.
 - ii) STOR demand and FR demand are assumed to be independent.
 - iii) The energy storage system always has enough energy to satisfy at least the demand from the STOR and the FR.

¹³ Short-term operating reserve is a grid ancillary service program to provide a contracted level of power when instructed by the national grid to meet the energy reserve requirement. Fast reserve is a grid ancillary service program to provide rapid and reliable delivery of active power through increasing output from generation or reducing consumption from demand sources.

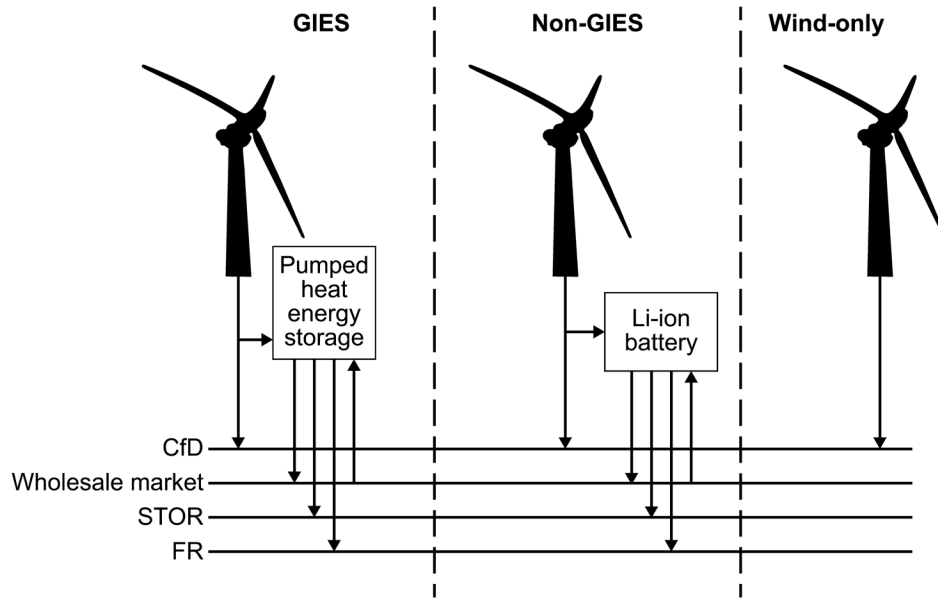


Figure A.1—Revenue sources with the generation and energy storage system; the arrows represent the energy flow

The technical, economic, and financial input data for the model are given in Table A.1, Table A.2, Table A.3, and Table A.4.

Table A.1—Project time, power rating, and efficiency parameters of GIES and non-GIES (technical specification)

Category	Input	GIES (thermal based)			Non-GIES (chemical based)		
		Min.	Most likely	Max.	Min.	Most likely	Max.
Project time	Construction time (years)	2	3	4	2	3	4
	System life (excluding construction) (years)	22	25	27	10	12	15
Power rating	Power rating for recovering energy from storage ($P_{\eta_{B4}}$) (MW)	$P_{\text{Har}} \times \eta_{B4}$					
	Power rating for putting energy into storage ($P_{\eta_{B3}}$) (MW)	$P_{\text{Har}} \times \eta_{B3}$					
	Power rating for electricity conversion ($P_{\eta_{B2}}$) (MW)	$P_{\text{Har}} \times \eta_{B2}$					
	Primary harvester power rating (P_{Har}) (MW)	100					
	Energy storage energy capacity (E_{ES}) (MWh)	100					
	Energy storage power capacity (P_{ES}) (MW)	50					
	Energy storage energy output at year 1 from wind energy ($E_{\text{ES-Har}}$) (MWh/year)	$E_{\text{Har}} \times \eta_X \times \beta_{\text{SO}}$					
	Total energy storage energy output at year 1 (wind + wholesale) ($E_{\text{ES-Output}}$) (MWh/year)	$E_{\text{STOR-Util}} + E_{\text{Sell-Wholesale}} + E_{\text{CfD}} + E_{\text{FR-Util}}$					
	Storage degradation (θ) (%/yr)	Negligible			2		
	Primary source energy output (E_{Har}) (MWh/year)	$P_{\text{Har}} \times CF \times 365 \times 24$					

Efficiency parameters	Capacity factor (CF) (%)	30					
	Power rating committed to STOR (P_{STOR}) (MW)	20					
	Storage (round-trip) efficiency (η_S) (%)	84.1	88.5	89	90.3	95.0	99.8
	Transmission efficiency (η_T) (%)	82.2	86.5	87	96.3	98.3	100.0
	Throughput efficiency (η_X) (%)	$\frac{\eta_T \times \eta_S}{\eta_S + (1 - \eta_S) \times \beta_{\text{SO}}}$					

Table A.2—Power ratios and revenue of GIES and non-GIES (technical specification)

Category	Input	GIES (thermal based)			Non-GIES (chemical based)		
		Min.	Most likely	Max.	Min.	Most likely	Max.
Power ratios	Fraction of electrical energy output from generator passed through energy storage (β_{SO}) (%)	17			15		
	Fraction of primary electrical energy input that will pass through energy storage (β_{SI}) (%)	$\frac{\beta_{\text{SO}}}{\eta_{\text{S}} + (\beta_{\text{SO}} \times (1 - \eta_{\text{S}}))}$					
	Power ratio for recovering energy from storage (η_{B4})	1					
	Power ratio for putting energy into storage (η_{B3})	1					
	Power ratio for electricity generation (η_{B2})	CF					
Revenue	Total hours availability commitment to STOR ($H_{\text{STOR-Avail}}$) (Hr/year)	3867					
	STOR: average utilization hours for northern region ($H_{\text{STOR-Util}}$) (Hr/year)	39.42					
	STOR: annual energy utilization ($E_{\text{STOR-Util}}$) (MWh/year)	$H_{\text{STOR-Util}} \times P_{\text{STOR}}$					
	ES energy for wholesale market ($E_{\text{Sell-Wholesale}}$) (MWh/year)	$\eta_{\text{S}} \times (E_{\text{ES-Har}} + E_{\text{Buy-Wholesale}}) - E_{\text{STOR-Util}} - E_{\text{FR-Util}}$					
	CfD energy: generator to grid (E_{CfD}) (MWh/year)	$(E_{\text{Har}} - E_{\text{ES-Har}}) \times \eta_{\text{T}}$					
	Cheap electricity purchase from wholesale ($E_{\text{Buy-Wholesale}}$) (MWh/year)	$\frac{E_{\text{ES}} \times 365 - E_{\text{ES-Har}}}{\eta_{\text{S}}}$					
	Fast reserve: total hours availability commitment ($H_{\text{FR-Avail}}$) (Hr/year)	448	2957.5	5040	448	2957.5	5040
	Fast Reserve: maximum energy utilization ($H_{\text{FR-Util}}$) (MWh/year)	0	422.5	1200	0	422.5	1200

Table A.3—Specific economic and financial specifications of GIES and non-GIES

Category	Index	Input	GIES (thermal based)			Non-GIES (chemical based)		
			Min.	Most likely	Max.	Min.	Most likely	Max.
Economic	F1	Specific fixed O&M power cost for generator ($C_{\text{O\&M-Gen}}$) (k£/MW-year)	30	45	74	53	59	67
	F2	Specific fixed O&M power cost for energy storage ($C_{\text{O\&M-Es}}$) (£/MW-year)	1.43×10^{-6}	2.2×10^{-6}	3.63×10^{-6}	1700	4750	7800
	F3	Specific generator overnight cost (C_{Har}) (£/MW)	832	1280	2112	1047	1164	1338
	F4	Specific balance of system (BOS) for generator cost ($C_{\text{BOP-Har}}$) (k£/MW)	249	384	633	296	349	419
	F5	Specific BOS for energy storage cost ($C_{\text{BOP-ES}}$) (k£/MWh)	0.83	2.80	4.77	117	139	166
	F6	Specific energy storage overnight cost (C_{ES}) (k£/MWh)	5.5	18.65	31.8	130	215	300
	F7	Overnight cost ($C_{\text{Overnight}}$) (£)	$E_{\text{ES}} \times (C_{\text{BOP-ES}} + C_{\text{ES}}) + P_{\text{Har}} \times C_{\text{Har}} + (\max(P_{\eta_{B3}}, P_{\eta_{B4}}) + P_{\eta_{B2}} \times C_{\text{BOP-Har}})$					
	F8	Annual inflation rate for cash (O&M and revenue) from 1998 to 2018 (%)	2.8					
Financial	G1	Cost of debt (K_D) (%)	4	5	6	3	4	5
	G2	Cost of equity (K_E) (%)	5	6	8	4	6	7
	G3	WACC (%)	$K_E \times \theta_{\text{CAPEX}} + K_D \times (1 - \theta_{\text{CAPEX}}) \times (1 - \theta_{\text{Tax}})$					
	G4	Escalation factor for construction costs (%)	0					
	G5	Depreciation factor for capital cost (%)	5					
	G6	Equity share on CAPEX (θ_{CAPEX}) (%)	30					
	G7	Effective tax rate (θ_{Tax}) (%)	11					
	G8	Interest earnings nominal rate (%)	5					

Table A.4—Economic specifications for revenue sources

Service	Input	Min.	Most likely	Max.
CfD	Strike price (£/MWh)	89.12	92.55	93.92
Wholesale market	Average daily expensive price (£/MWh)	62.00	71.77	83.15
	Average daily inexpensive price (£/MWh)	20.00	35.73	40.91
STOR	Average availability hours price (£/MW/hr)	4.25		
	Average utilization hours price (£/MWh)	150.57		
Fast reserve	Availability hours price (£/hr)	160.00	277.75	504.00
	Utilization hours price (£/MWh)	84.00	97.875	106.00

The risk analysis gives probability distribution functions (PDFs) for net present value (NPV), levelized cost of electricity (LCOE), and internal rate of return (IRR). This is particularly useful when there are model uncertainties.

The risk analysis is conducted with Monte Carlo analysis with random sampling based on the probability distribution defined for each input. The Monte Carlo stopping criteria is based on the convergence requirement, with a convergence tolerance at 1% and a confidence level of 95%. More than 50 000 iterations are performed.

Being a novel technology, the technical, economic, and financing uncertainties are higher for GIES. Consequently, as expected, in Figure A.2 there is a wider LCOE uncertainty for GIES as it is currently in the study and feasibility stage, whereas non-GIES, being in bid/tender stage, has more cost certainty. For wind-only, the LCOE and its uncertainty are the lowest as the technology is mature.

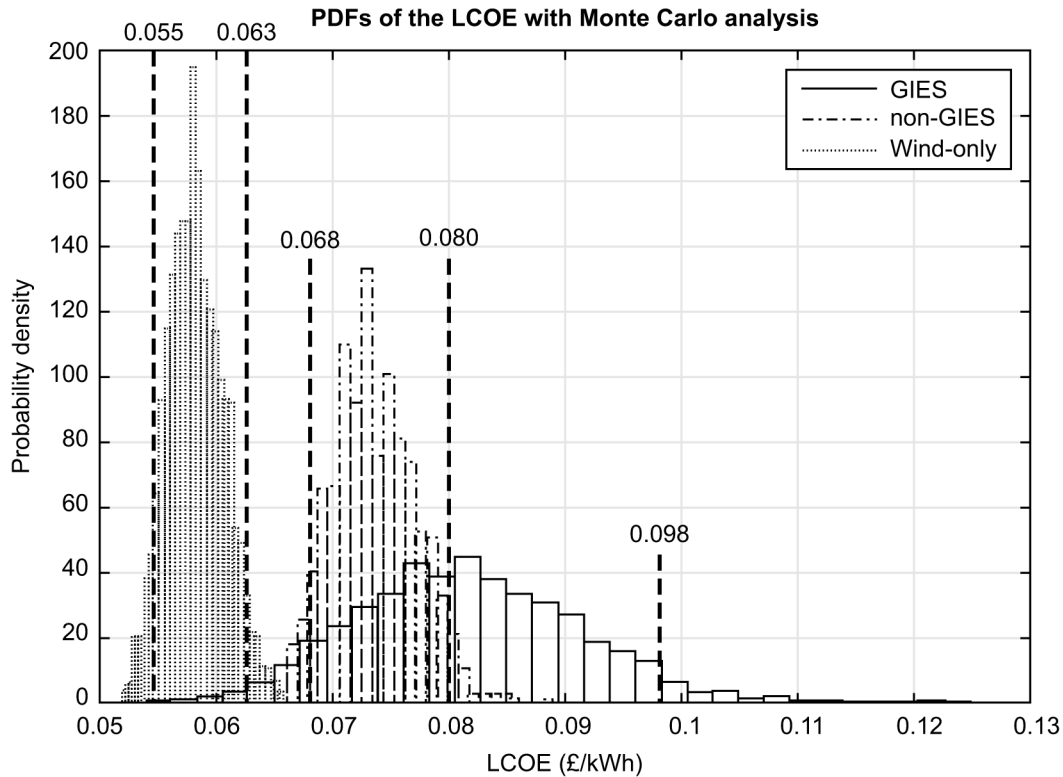


Figure A.2—PDFs of the LCOE with MC analysis

Figure A.3 presents the PDFs of the IRR for GIES and non-GIES. As the IRR is the discount rate when the NPV of the cash flow is equal to zero, a negative IRR implies that the initial investment is greater than the discounted cumulated cash flow for the operations.

IRR is a relative measure. For sense-making, it is necessary to refer to relevant values. For energy infrastructure, the European Commission Benchmark suggests a social discount rate (the WACC used to reflect the social view on how potential costs and benefits can be valued against present ones) of 5% (European Commission [B3]). The International Renewable Energy Agency (IRENA) mentioned that the WACC for renewable power generation for Organisation for Economic Co-operation and Development (OECD) countries and China is 7.5% (IRENA [B7]). Grant Thornton [B5] reports the WACC to be at 8% for onshore wind. Cavazzi and Dutton [B1] set the WACC for offshore wind energy to be at 10%.

The IRR for wind-only is the highest compared to GIES and non-GIES. This is due to the attractive revenue sources, in particular the CfD. For GIES, the probabilities for the IRR to be greater than 5% and 10% are 99.4% and 75.5%, respectively. For non-GIES, the probabilities for the IRR to be greater than 5% and 10% are 99.9% and 5.7%, respectively. Based on the above discussions, wind-only is a more profitable investment. The IRRs for GIES and non-GIES are in the region of the WACCs, as discussed by government, industries, and academics.

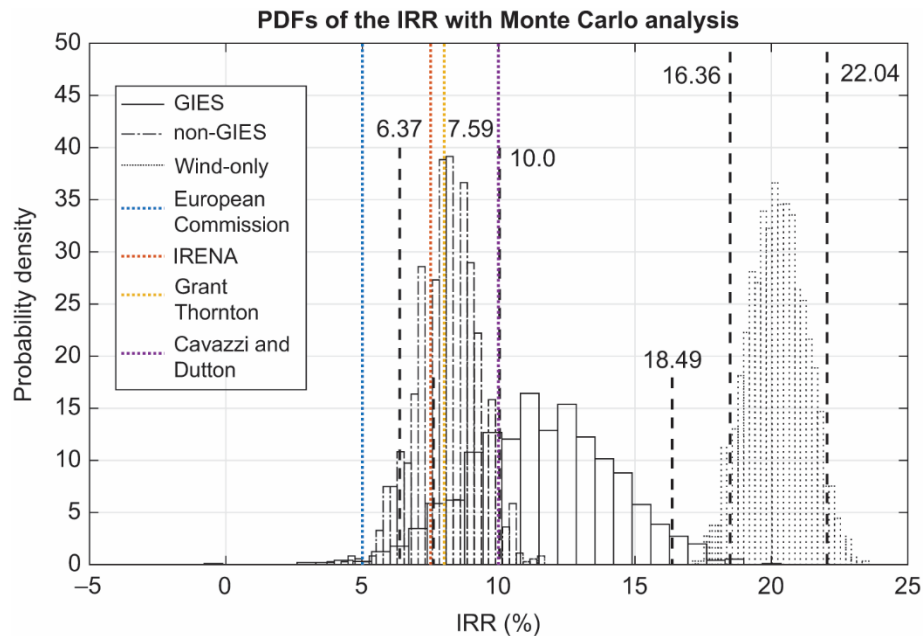


Figure A.3—PDFs of the IRR with MC analysis

Figure A.4 presents the PDFs of the NPV to equity for GIES and non-GIES. For wind-only, the NPV is always in the positive region. This is confirmed by the popularity of wind-only investments in the last decade. As supported by the deterministic analysis, the variance is higher for GIES, as there is a high variance for the system's cost. The mean NPV for non-GIES is lower than GIES but with less variance. The probabilities for the NPV to be greater than zero for GIES and non-GIES are 98.4% and 98.0%, respectively.

In summary, the investment in a wind farm alone is the most profitable and less risky. This is consistent with the plethora of this kind of plant developed all over the world. Energy storage systems (GIES and non-GIES) are broadly comparable and present more risk and less return than a wind farm.

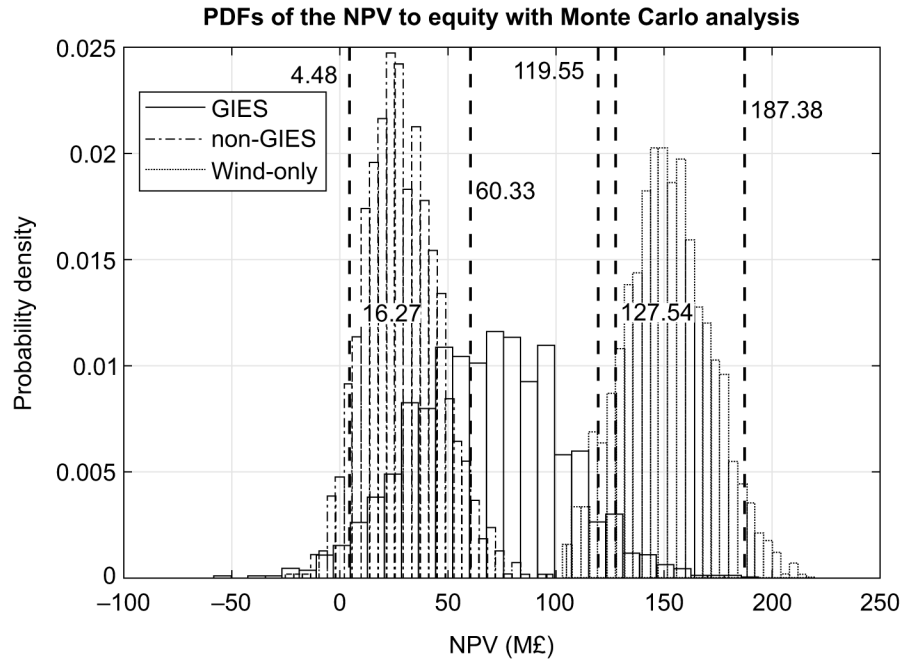


Figure A.4—PDFs of the NPV to equity with Monte Carlo analysis

A.2 Levelized cost of energy analysis for solar photovoltaic power generator with vanadium redox battery and Li-ion energy storage systems in Africa

The following example is reproduced from Lai and McCulloch [B13].

A scenario has been developed with the load and generation data from two African countries. The national load curve has been downsized with the peak load at 2 MW to represent a load demand for a typical community in Africa. The rated capacity of the photovoltaic system is 5 MW, with the energy storage capacity to be rated at 3 MWh. The calculated capacity factors for the photovoltaic system are 12.02%, 11.67%, 12.26%, and 12.61% for years 2009, 2010, 2011, and 2012, respectively. The purpose of this example is to present the $LCOE_{\text{system}}$ and levelized cost of delivery (LCOD) for a real-life renewable energy system with energy storage systems. $LCOE_{\text{system}}$ is the LCOE for the combined assets, photovoltaic and energy storage system. Two types of dominant energy storage system technologies, Li-ion and vanadium redox battery (VRB), are studied. In this study, three ranges of costs values are used to determine the cost of electricity for LCOD and $LCOE_{\text{system}}$. These are the lower-bound, upper-bound, and medium-bound cost, as shown in Table A.5. The cost calculation is performed for each year of the four years of data. Sensitivity analysis is conducted for a range of discount rates. One of the vital input parameters for LCOE calculations is the value of the discount rate. The discount rate essentially considers the time value of money, through monetary depreciation as well as the investment risks. Photovoltaic systems are considered much higher risk when compared with traditional power plants, and therefore, they will be at higher discount rates.

Table A.5—Cost and technical specification of the system components

Inputs	Photovoltaic system	Energy storage system	
		VRB	Li-ion battery
Capital cost (C_{Cap})	120 (\$/unit)	760 to 1600 (\$/kWh)	715 to 1640 (\$/kWh)
Installation cost (C_{Inst})	108 (\$/unit)	N/A	N/A
O&M cost ($C_{O\&M}$) $\times$ (\$/unit/year) $\times$ 100	(\$/unit/year)	100 to 140 (\$/kWh)	80 to 95 (\$/kWh)
System lifetime (n)	N/A	20 years	15 years
Round-trip efficiency (η)	N/A	70%	90%
Degradation rate	0.5%/year	0.01%/year	2%/year

The results are summarized in the form of a box plot in Figure A.5 and Figure A.6. Figure A.5 shows that LCOD is higher with Li-ion than with vanadium redox battery for both the median and minimum value when the discount rate is less than 8%. The discount rate needs to be 10% or more when the LCOD for vanadium redox battery is higher than that for Li-ion.

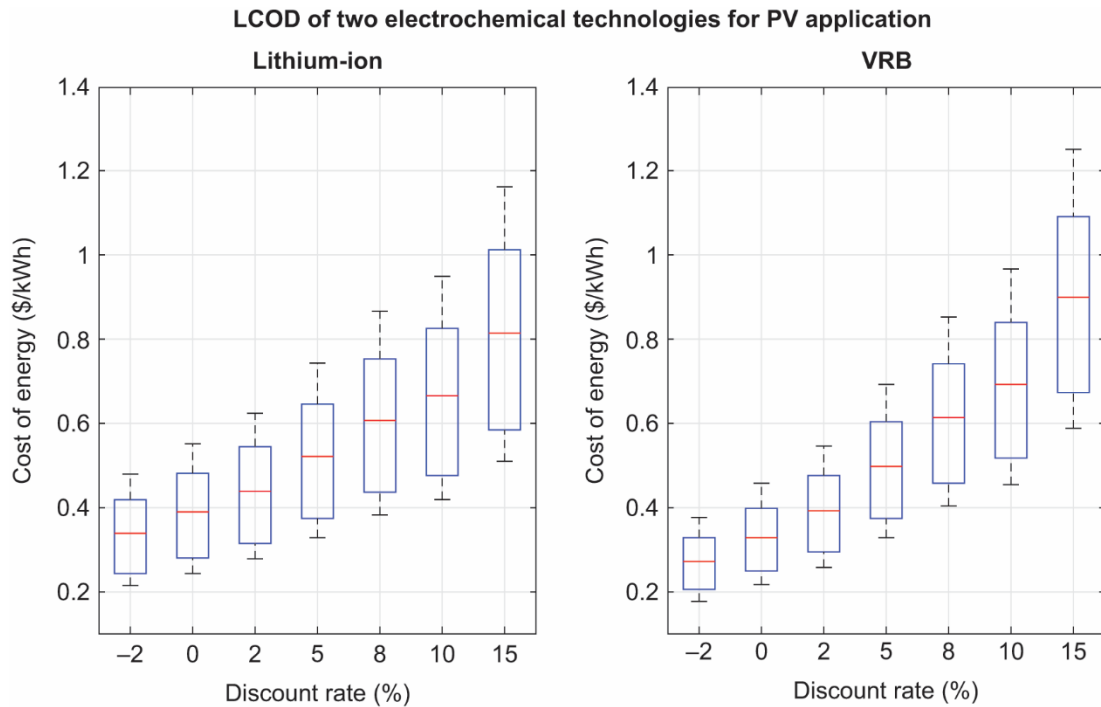


Figure A.5—LCOD for Li-ion and VRB at various discount rates

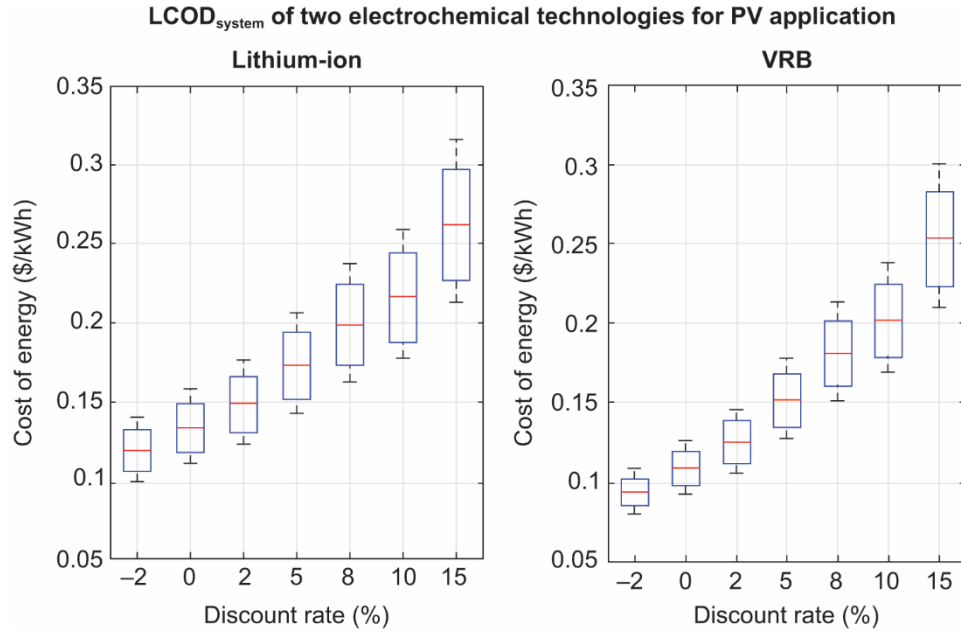


Figure A.6—LCOE_{system} for Li-ion and VRB at various discount rates

The results for LCOE_{system} at various discount rates for the two energy storage technologies are given in Figure A.6. The LCOE is lower for VRB in all the studied discount rates and price range considerations. Although Li-ion battery has a higher round-trip efficiency than VRB, the lifetime of the system is considerably shorter, and this significantly affects the LCOE_{system}.

From Figure A.5 and Figure A.6, it could be understood that adding energy storage to the system can increase the LCOE_{system} at a high discount rate. Also, the cost variation is wider. This is due to the capital and installation cost of the system being more dominant. At a low discount rate, the value of energy will have less depreciation with respect to time.

Annex B

(informative)

Repository data

Table B.1—Overnight and operating costs for five energy storage (ES) technologies (Lai and Locatelli “Economic and financial” [B10])

	Type	ES	Overnight cost (£/kWh)	Fixed O&M cost (£/kW-year)
Generation integrated energy storage (GIES)	Mechanical	Pumped storage hydropower (PSH)	3.8 to 38.2	2.3 to 7.8
		Compressed air energy storage (CAES)	5.5 to 31.8	1.6 to 10.6
		Flywheel	339.5 to 733	3.7 to 5.1
	Heat	Pumped thermal energy storage (PTES)	11 to 17.8	0.0022
Non-GIES	Electro-chemical	Lithium-ion	130 to 300	1.7 to 7.8
		Vanadium redox battery (VRB)	113 to 650	2.6 to 12.6

Table B.2—Generic cost breakdown of four energy storage systems (International Renewable Energy Agency “Electricity storage” [B6])

Type	ES technology	Percentage of capital cost (%)
GIES (mechanical)	Pumped storage hydropower (PSH)	Powerhouse: 37; upper reservoir: 19; tunnels: 6; powerhouse excavation: 4; EPC and management: 17; and owner’s costs: 17
	Compressed air energy storage (CAES)	Cavern: 40, turbine: 30, compressor: 14, balance of system (BOS): 7; EPC and management: 3; and owner’s costs: 6
Non-GIES (electro-chemical)	Lithium-ion	Materials costs (e.g., anode, cathode, separator, and electrolyte): 30 to 60; labor: 5 to 20; and overhead: 5 to 30
	VRB	Peripherals (fluid regulation: 4, assembling: 9, control engineering: 9, and power electronics: 9); electrolyte and tanks: 35; and stack: 34

Annex C

(informative)

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Bibliographical references are resources that provide additional or helpful material but do not need to be understood or used to implement this standard. Reference to these resources is made for informational use only.

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¹⁴ EIA publications are available from the U.S. Energy Information Administration (<https://www.eia.gov/>).

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